

# The Institutional Flow Clock

A pre-registered test of trading against calendar-scheduled institutional flows

Minh Hoang, University of Florida · Gator Quant Hacks 2026, Systematic Trading track

Code and full audit trail: <https://github.com/minh-stakc/gqh-flow-clock>

## 1. Summary

Pension, target-date and balanced funds, index-benchmarked bond managers and institutions raising month-end cash trade on calendars fixed by mandates, index rules and settlement plumbing. We pre-registered (git commit before any backtest) three sleeves that take the other side of these flows, a scale-free path-geometry trend variant, and a rule for choosing the submission. A logged, overfitting-tested ensemble search (1,404 configurations, probability of backtest overfitting 0.023) selected **rebalancing pressure (A) + Treasury month-end index extension (C) + time-series momentum**, which earned an in-sample (2006–2024) Sharpe ratio of **0.98** net of costs (0.62 at 2x costs), a  $-11.7\%$  maximum drawdown and  $6.2\%$  annual alpha to FF5 + momentum + bond + commodity factors ( $t = 3.95$ ). An independent replication on CME futures (Databento) gave 1.05 for the flow sleeves. **The single out-of-sample run (Oct 2024–Oct 2026) returned  $-0.12$**  (0.38 before costs), the bottom 1% of the strategy’s own in-sample two-year windows. Both mechanisms kept their sign but shrank to roughly 25–60% of their in-sample size, too little to pay for trading. We report this as a well-tested negative result and explain what we would change.

## 2. Economic hypothesis: who is on the other side?

| Sleeve                      | Counterparty and why they cannot stop                                                                                                                                                   | Rule (decide at close, fill next close)                                                                                    | Prediction / kill condition                                    |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| A. Rebalancing pressure     | 60/40, target-date and pension funds must sell the asset that outperformed back to fixed weights at month end (Harvey, Mazzoleni & Melone 2025); policy is set by investment committees | Drift of a 60/40 SPY/IEF portfolio at $T-6$ , standardised; hold $-$ drift in SPY vs IEF over $[T-4, T]$ at 5% vol per leg | SPY–IEF over the window opposes the drift                      |
| B. Dash for cash            | Institutions must sell to have cash on payment dates; the last trade that settles by month end is $T-k$ (Etula et al. 2020)                                                             | Short SPY over $[T-k-5, T-k-1]$ , long over $[T-k, T+3]$ ; $k = 3/2/1$ under $T+3/T+2/T+1$                                 | Windows shift with each settlement change (natural experiment) |
| C. Treasury index extension | Bond indices add new issues only at month end; benchmarked managers must buy duration into it (Hartley & Schwarz 2019)                                                                  | Long IEF over $[T-2, T]$ , sized by that month’s announced coupon issuance (DV01-weighted)                                 | IEF excess return over the window $> 0$                        |
| Trend leg                   | Investors under-react to slow information (Moskowitz, Ooi & Pedersen 2012)                                                                                                              | 12-month sign, vol-scaled, 15 ETFs                                                                                         | Diversifies the contrarian flow sleeves                        |

The edge persists because the counterparties are price-insensitive by mandate: changing a rebalancing policy, an index rule or the settlement cycle requires governance, not a trading decision. What would kill it: crowding once the patterns are published, flows that move earlier, or costs larger than the effect. We also pre-registered a novel trend statistic extending the author’s earlier work:

the bridge-excursion ratio  $R = A/(L\sqrt{W})$  (mean absolute deviation of log prices from the chord over mean absolute daily return), which converges to  $\pi/8$  under a random walk and is invariant to volatility, used to scale trend positions by path persistence (PCT).

### 3. Data and universe

Daily total-return-adjusted OHLCV for US ETFs (Yahoo Finance; splits and distributions folded into prices), 3-month T-bill (FRED DTB3) as the cash rate, Fama-French 5 factors and momentum (K. French library, available to Aug 2026), and every Treasury auction since 2002 with its announcement date (U.S. Treasury Fiscal Data). Replication: CME futures from Databento GLBX.MDP3 (ES, ZN and 20 markets), volume-ranked front contracts with returns always computed within one contract (no roll gaps), plus one round trip per roll; ES/ZN sampled at 16:00 ET from hourly bars so they align with ETF closes (daily correlation with SPY/IEF 0.998/0.987). History 2005-01 to 2026-10-02. **Out-of-sample:** the brief’s 20%-or-2-years rule gives 2024-10-03 to 2026-10-02 (501 days). *Survivorship:* instruments are broad, still-listed ETFs that launched before the sample (no delisting risk) and futures; *corporate actions:* total-return adjustment; *missing data:* never filled forward across more than an isolated print; unscheduled exchange closures (e.g. Hurricane Sandy, 2012) are treated as unknown until announced (an audit found and we fixed a one-month lookahead from using the realised calendar).

### 4. Methodology

**Protocol.** (i) Hypotheses, windows, signs, sizing, variant neighbourhoods, the selection rule and three gates were committed before the first backtest; (ii) an amendment adding a diversified ensemble search with overfitting controls was committed before any in-sample result was read; (iii) the selection was frozen in `results/selection.json` and committed; (iv) the out-of-sample window, which the data loaders refuse to return unless explicitly unlocked, was evaluated once (one timestamped pass in `results/oos_log.csv`). Every backtest ever run is appended to `results/trials.csv`: 1,467 in-sample trials.

**Execution and costs.** Signals use data up to the close of day  $d$  and fill at the close of  $d+1$  (ETF trend: next open). One-way costs: 3 bp for SPY/IEF-class ETFs, 5 bp sector/commodity/currency ETFs, 10 bp otherwise, plus 30 bp/yr to borrow shorts; futures 1 bp (ES, ZN) to 5 bp; cash earns the T-bill. These are several times quoted half-spreads (SPY and IEF trade at 1 cent on \$100–\$600) to cover slippage. Every number is net; every headline is repeated at 2x costs.

**Ensemble and selection.** Streams: sleeves A–E (D: FX hedge flows, E: auction cycle, exploratory) and the trend leg (PCT or plain TSMOM). The search covers stream subsets, three weighting schemes (equal risk, shrunk inverse-volatility, Ledoit-Wolf minimum variance; all from expanding windows of past returns, lagged), volatility targets of 6/8/10% and an optional drawdown brake. Rule: highest in-sample Deflated Sharpe ratio (Bailey & López de Prado 2014) among configurations passing G1 (Sharpe  $> 0$  at 2x costs), G2 (no year  $> 40\%$  of P&L) and G3 (median Sharpe of one-step neighbours  $\geq$  half its own), provided the CSCV probability of backtest overfitting (Bailey et al. 2017; 16 blocks, 12,870 splits) is below 0.25.

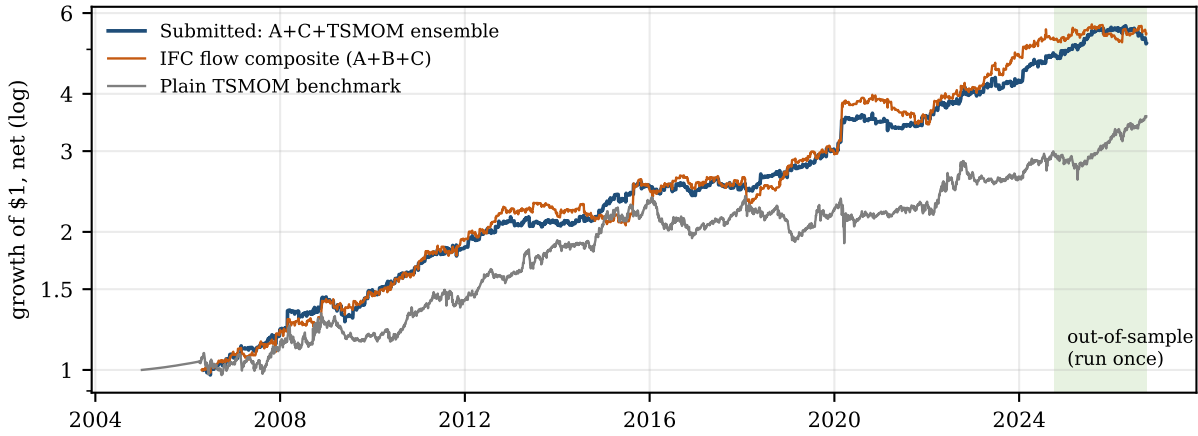
**Verification.** Each strategy module was implemented by one agent and attacked by an independent reviewer that searched for lookahead (including random-walk perturbation of all data after cut dates), spec drift and bugs, and re-derived headline numbers with separate code. Two bugs were found and fixed before selection: the closure-calendar lookahead above, and an ensemble overlay

that charged re-scaling costs on idle streams (detected from an impossible  $-124\%$  day); the full search was re-run.

## 5. Results

**Table 1:** Submitted strategy (A + C + TSMOM, minimum-variance weights, 8% vol target). Net of costs except “Gross”; “2x costs” and “Gross” are Sharpe ratios; turnover is total trades/NAV per year.

|                       | Ann. ret | Vol  | Sharpe       | 2x costs | Gross | Max DD | Worst mo. | Turnover | NW $t$ |
|-----------------------|----------|------|--------------|----------|-------|--------|-----------|----------|--------|
| In-sample 2006–24     | 9.0%     | 7.6% | <b>0.98</b>  | 0.62     | 1.35  | −11.7% | −4.7%     | 90×      | 4.62   |
| Out-of-sample 2024–26 | 3.0%     | 6.6% | <b>−0.12</b> | −0.61    | 0.38  | −8.7%  | −2.3%     | 108×     | −0.18  |



**Figure 1:** Growth of \$1, net of costs. The shaded period was evaluated once, after the selection was committed.

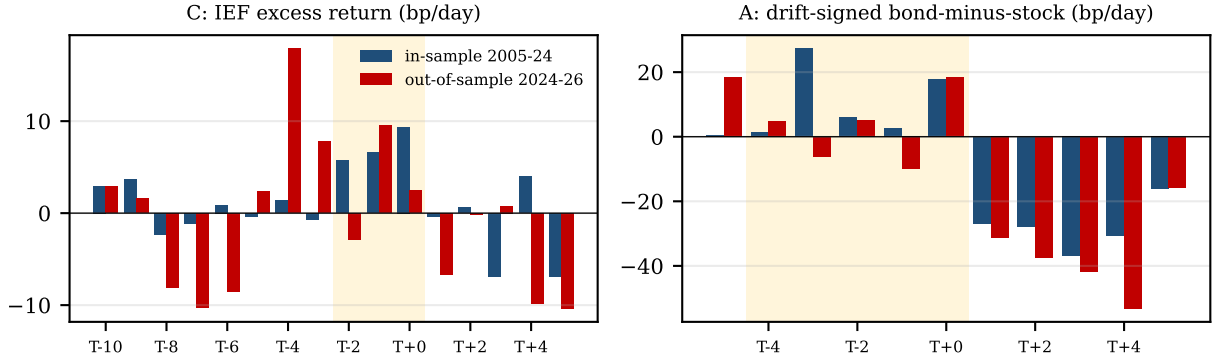
**Table 2:** Every pre-registered stream and replication: net Sharpe ratio. Futures in-sample starts 2010-06.

| ETF streams                   | IS    | OOS   | OOS 2x | Composites / futures       | IS   | OOS   |
|-------------------------------|-------|-------|--------|----------------------------|------|-------|
| A rebalancing pressure        | 0.47  | −0.73 | −1.01  | IFC flow composite A+B+C   | 0.94 | −0.31 |
| B dash for cash               | 0.19  | −0.27 | −0.42  | IFC-F on ES/ZN (Databento) | 1.05 | 0.01  |
| C Treasury month end          | 0.77  | −0.09 | −0.45  | PCT-F, 20 futures          | 0.24 | −0.10 |
| D FX hedge (exploratory)      | −0.68 | 0.28  | −0.37  | TSMOM-F, 20 futures        | 0.28 | 0.14  |
| E auction cycle (exploratory) | 0.16  | −1.06 | −1.79  | ID-conditioned trend-F     | 0.43 | 0.06  |
| PCT (path-geometry trend)     | 0.30  | 0.71  | 0.60   |                            |      |       |
| TSMOM benchmark               | 0.44  | 0.55  | 0.49   |                            |      |       |
| ID-conditioned trend          | 0.57  | 0.49  | 0.43   |                            |      |       |

**In-sample evidence.** Sharpe by sub-period: 1.29 (2006–08), 1.23 (2009–12), 0.57 (2013–16), 1.09 (2017–20), 0.79 (2021–24); best year 11% of P&L; positive in 2008 and 2022; 90% block-bootstrap interval  $[0.67, 1.31]$ . Factor loadings are small (market 0.05, momentum 0.05, Treasury duration 0.31). One-step neighbours have median Sharpe 0.94; the median of all 1,404 configurations is 0.33, so diversification, not a lucky configuration, drives the result (PBO 0.023). With all 1,467 trials the expected maximum Sharpe under no skill is 1.04 and the Deflated Sharpe is 0.40; the trials are so correlated that their effective number is 3.5 (eigenvalue participation), giving 0.999. The flow composite (A+B+C) has in-sample alpha 6.5% ( $t = 3.11$ ), but timing matters: shifting every window one day earlier/later gives 0.52/0.53, and one extra day of delay gives 0.59.

**What failed in-sample.** The settlement natural experiment: the old fixed T+3 windows did as well as the settlement-aware ones (0.23 vs 0.19), so B2 is rejected. Issuance sizing added nothing to C (0.78 without it). PCT did *not* beat plain TSMOM (0.30 vs 0.44), on ETFs or futures (0.24 vs 0.28): its kill condition triggered. The information-discreteness variant (Da, Gurun & Warachka 2014) did improve trend on both universes (0.57, 0.43). FX hedge flows and the auction cycle failed, as the decay literature predicted for the latter.

**Out-of-sample.** Gross of costs the strategy still earned a Sharpe of 0.38; costs on  $108\times$  annual turnover took it to  $-0.12$ . Contributions: A  $-1.3\%$ , C  $-0.2\%$ , trend  $1.1\%$  a year. Raw effects (Figure 2, no strategy code): IEF earned 3.0 bp/day over  $[T-2, T]$  versus  $-1.6$  on other days (in-sample 7.2 vs  $-0.4$ ); the drift-signed bond-minus-stock spread over  $[T-4, T]$  fell from 11.0 to 2.5 bp/day. The rebalancing paper became public in March 2025, inside the window, consistent with post-publication decay (McLean & Pontiff 2016). The strategy’s in-sample two-year rolling Sharpe had median 0.97, 5th percentile 0.17 and minimum  $-0.29$ : the out-of-sample result is unusually poor, not ordinary noise. PCT, rejected in-sample, was the best out-of-sample stream (0.71); we do not switch to it, since that would be selection on the test set.



**Figure 2:** Mean daily excess return by trading day relative to month end  $T$ ; shaded: traded window. Right panel signs the bond-minus-stock spread by the 60/40 drift observed at  $T-6$ ; the post-turn reversal is what temporary price pressure predicts.

## 6. Risk management

**Limits.** Gross exposure  $\leq 4\times$  and  $\leq 3\times$  per ETF (binding on 12% of days, the month-end windows); average gross  $0.87\times$ . Each stream is vol-scaled at entry; the portfolio targets 8% volatility from a lagged 63-day EWMA, so exposure falls automatically after volatility spikes. The cap bounds single-window losses: a 3-day  $4\times$  IEF window loses about 2.8% per one-sigma rate move (5.7% at two sigma); the worst in-sample month was  $-4.7\%$ . **Pre-set de-risking.** The pre-registered drawdown brake (halve exposure 10% below peak) never triggered in-sample at 1x costs; we would run it live, plus a stop on each mechanism if its rolling 24-month raw effect (Figure 2) turns negative, which the out-of-sample data would have tripped for A. **Factor and regime exposure.** Market and momentum betas are near zero; the duration beta (0.31) comes from sleeve C and is the main macro risk (it is a long-duration bet three days a month). The strategy made money in 2008, 2020 and 2022 (positive stock-bond correlation). Daily skew is positive (1.04): the contrarian legs are not short volatility. **Event risk.** CPI, payrolls or FOMC inside a month-end window can dominate a 3–5 day position; live, we would skip or halve windows containing a scheduled top-tier release.

## 7. Liquidity and capacity

Square-root impact,  $\text{cost} = \text{fixed} + \sigma_{\text{daily}} \sqrt{\text{trade}/\text{ADV}}$ , using each instrument’s trailing 63-day median dollar volume and the strategy’s actual trades. Net Sharpe by assets under management:

| AUM                              | \$0 (fixed costs only) | \$1M | \$10M | \$100M | \$1B  |
|----------------------------------|------------------------|------|-------|--------|-------|
| ETF ensemble (2019–24 liquidity) | 0.71                   | 0.49 | −0.01 | −1.42  | −3.57 |
| Futures, IFC-F (2010–24)         | 0.99                   | 0.98 | 0.94  | 0.83   | 0.47  |

The ETF version concentrates up to  $4\times$  NAV of IEF (secondary volume about \$0.5bn a day) into month-end windows, so it is a 20-year proof of the effect, not a vehicle: its edge halves near \$1–2M. This is conservative, since ETF screen volume understates liquidity reachable through creation/redemption against the Treasury cash market. ZN and ES trade \$50–300bn a day, so the futures implementation keeps half its edge near \$1bn. Turnover is high ( $90\times/\text{yr}$ ), which is why costs, not signal, decided the out-of-sample result.

## 8. Limitations and next steps

The out-of-sample window holds only 24 month-ends, so a two-year Sharpe has a standard error near 0.7; still, the result sits at the bottom of the in-sample distribution and both mechanisms weakened, so we read it as decay, not bad luck. Daily bars cannot time the 4 pm close or intraday flow; ETF adjusted prices are revised retroactively (we snapshot them); the variant count, though fully logged, is large. Next: (1) trade only the futures version with market-on-close execution and monitor each mechanism’s raw effect live; (2) intraday Databento data to locate when month-end flows now hit (the OOS profile suggests earlier and after the turn); (3) treat the information-discreteness trend result, which replicated across ETF and futures universes, as a new pre-registered hypothesis rather than a post-hoc fix.

## References

Bailey, D., López de Prado, M. (2014), The Deflated Sharpe Ratio, *J. Portfolio Management*. Bailey, D., Borwein, J., López de Prado, M., Zhu, Q. (2017), The Probability of Backtest Overfitting, *J. Computational Finance*. Da, Z., Gurun, U., Warachka, M. (2014), Frog in the Pan, *Review of Financial Studies*. Etula, E., Rinne, K., Suominen, M., Vaitinen, L. (2020), Dash for Cash, *Review of Financial Studies*. Hartley, J., Schwarz, K. (2019), Predictable End-of-Month Treasury Returns, working paper. Harvey, C., Mazzoleni, M., Melone, A. (2025), The Unintended Consequences of Rebalancing, NBER WP 33554. Harvey, C., Liu, Y., Zhu, H. (2016), ...and the Cross-Section of Expected Returns, *RFS*. Ledoit, O., Wolf, M. (2004), A well-conditioned estimator for large covariance matrices, *J. Multivariate Analysis*. Lou, D., Yan, H., Zhang, J. (2013), Anticipated and Repeated Shocks in Liquid Markets, *RFS*. McLean, R., Pontiff, J. (2016), Does Academic Research Destroy Stock Return Predictability?, *J. Finance*. Melvin, M., Prins, J. (2015), Equity Hedging and Exchange Rates at the London 4 p.m. Fix, *J. Financial Markets*. Moskowitz, T., Ooi, Y., Pedersen, L. (2012), Time Series Momentum, *J. Financial Economics*. Almgren, R., Thum, C., Hauptmann, E., Li, H. (2005), Direct Estimation of Equity Market Impact, *Risk*. Data: Yahoo Finance; FRED; Kenneth R. French Data Library; U.S. Treasury Fiscal Data; Databento (GLBX.MDP3). Prior work extended: [github.com/minh-stakc/physics-quant-research](https://github.com/minh-stakc/physics-quant-research). Code was written with AI assistance; all results reproduce from the repository with `run_all.py`.